Xiaoyu Zhang

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Research Summary

My research focuses on the convergence of **wireless sensing**, **computational imaging**, and **smart health**. I turn low-cost commodity mmWave radar into a high-resolution imaging instrument for the human body, and the system built in my dissertation now images wounds *through their dressings* at a vascular surgery clinic under an IRB-approved protocol.

17 peer-reviewed papers (7 first-authored), **4 filed patents** (one under commercialization by a medical-device startup), and a **Best Paper Candidate Award**.

Research Interests

I build sensing systems that reach real patients and real sites, along three lines:

- **mmWave sensing of the human body:** wound moisture and area imaging through dressings [IoTJ'25] [HumanSys'25], a deployed clinical station with longitudinal patient scanning, pixel-level hand localization from a single commodity radar [IoTJ'25], mmWave–voice identification [UIST'23], deployment-aware evaluation of COTS mmWave radar for privacy-aware human sensing [S3'26], and a survey of mmWave for medicine [Sensors'25].
- **Multimodal medical AI:** chronic foot-ulcer prognosis from photoacoustic imaging [npj Imaging'26] [Photoacoustics'25], Alzheimer's screening from speech [ICIBM'26] [JBHI'24].
- **Sensing for sustainability:** robotic subsurface inspection of building thermal insulation [IROS'26].

Current Position

University at Buffalo, the State University of New York (SUNY) *Sep. 2026 - Present*
Postdoctoral Associate, Computer Science and Engineering Supervisor: Prof. Wenyao Xu

Education

University at Buffalo, the State University of New York (SUNY) *Sep. 2021 - Jul. 2026*
Ph.D., Computer Science and Engineering Co-advised by Prof. Wenyao Xu & Yaxiong Xie
Dissertation (**defended Jul. 2026**): *Toward High-Resolution and High-Capacity mmWave Sensing Technology for Medical Applications*
Committee: Wenyao Xu, Yaxiong Xie, Jun Xia

University of Science and Technology of China *Sep. 2017 - Jun. 2020*
M.S., Electronics and Communication Engineering Advised by Prof. Bin Liu

Hefei University of Technology *Sep. 2013 - Sep. 2017*
B.Eng., Electronic Information Engineering

Publications

Accepted / published: 17 papers, 7 first-authored (name in **bold**). First-author work appears at IEEE IoTJ (CAS Tier-1 TOP), IEEE/RSJ IROS and ICIBM; co-authored work at ACM IMWUT, ACM UIST, npj Imaging, Photoacoustics and IEEE JBHI. Tags: [Wireless Sensing] [Medical AI] [Sustainability] .

[**HumanSys'26**] [Medical AI] Henry Yang, Emily Liu, **Xiaoyu Zhang**, Wei Bo, Yutong Wu, Yike Liu, Qun Xie, Wenyao Xu, "*A Real-World Study of Sport-Based Support for Autism Spectrum Disorder*", The 4th International Workshop on Human-Centered Sensing, Modeling, and Intelligent Systems (HumanSys workshop, co-located with ACM MobiCom), Austin, USA, 2026.

[**S3'26**] [Wireless Sensing] Kirtirajsinh Parmar, Liyao Li, **Xiaoyu Zhang**, Wei Bo, Yaxiong Xie, Wenyao Xu, "*Towards Deployment-Aware Evaluation of COTS mmWave Radar for Privacy-Aware Surveillance*", ACM Workshop on Wireless of the Students, by the Students, and for the Students (S3, co-located with ACM MobiCom), Austin, USA, 2026.

[**IROS'26**] [Sustainability] **Xiaoyu Zhang**, Ye Zhan, Chenhan Xu, Zhengxiong Li, Chuqin Huang, Yanda Cheng, Licheng Liang, Wei Bo, Jun Xia, Chi Zhou, Hongyue Sun, Wenyao Xu, "*Autonomous Robotic Inspection of Building Thermal Insulation via Subsurface mmWave Sensing*", IEEE/RSJ International Conference on Intelligent Robots and Systems, 2026. *(Patent filed.)*

[**IMWUT'26**] [Medical AI] Shuwei Hou, Wei Bo, L. Cao, Varun Shijo, Chuhui Liu, **Xiaoyu Zhang**, L. Guo, Wenyao Xu, "*Speech Annotation and Transcription Enhancer (SATE): An Automated System for Child Language Sample Analysis*", Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 2026.

[**ICIBM'26**] [Medical AI] **Xiaoyu Zhang**, Wei Bo, Anarghya Das, Longxiang Pan, Wenyao Xu, "*Matching Speech Models to ADRD Subtypes: A Mechanistic Benchmark*", International Conference on Intelligent Biology and Medicine, 2026.

[**npj Imaging'26**] [Medical AI] Yanda Cheng, Chuqin Huang, Shuliang Yu, Saptarshi Chakraborty, Yunqi Xi, Robert W. Bing, Huijuan Zhang, **Xiaoyu Zhang**, Isabel Komornicki, Linda M. Harris, Wenyao Xu, Jun Xia, "*Predictive modeling of chronic foot ulcer outcomes using longitudinal photoacoustic imaging*", npj Imaging, 2026.

[**HealthCom'26**] [Medical AI] Nicholas Wei, Austin Xu, **Xiaoyu Zhang**, "*Pool or Specialize? Anatomic-Site Domain Shift in Melanoma Classification*", IEEE International Conference on E-health Networking, Application & Services, 2026.

[**IoTJ'25**] [Wireless Sensing] **Xiaoyu Zhang**, Zhengxiong Li, Yanda Cheng, Chenhan Xu, Chuqin Huang, Emma Zhang, Ye Zhan, Wei Bo, Jun Xia, Wenyao Xu, "*mmSkin: An Over-gauze Wound Assessment System using Radio Frequency Technologies*", IEEE Internet of Things Journal, 2025. *(Translated into US patent; under commercialization by a medical-device startup.)*

[**IoTJ'25**] [Wireless Sensing] **Xiaoyu Zhang**, Zhengxiong Li, Chenhan Xu, Luchuan Song, Huining Li, Hongfei Xue, Yingxiao Wu, Wenyao Xu, "*mmHand: Towards Pixel-Level-Accuracy Hand Localization Using a Single Commodity mmWave Device*", IEEE Internet of Things Journal, 2025.

[**HumanSys'25**] [Wireless Sensing] **Xiaoyu Zhang**, Zhengxiong Li, Yanda Cheng, Chenhan Xu, Chuqin Huang, Emma Zhang, Ye Zhan, Wei Bo, Jun Xia, Wenyao Xu, "*Through-dressing Wound Monitoring Based on the mmWave Sensor*", The 3rd International Workshop on Human-Centered Sensing, Modeling, and Intelligent Systems (HumanSys workshop, co-located with ACM SenSys), 2025. (**Best Paper Candidate**)

[**Sensors'25**] [Wireless Sensing] **Xiaoyu Zhang**, Chuhui Liu, Yanda Cheng, Zhengxiong Li, Chenhan Xu, Chuqin Huang, Ye Zhan, Wei Bo, Jun Xia, Wenyao Xu, "*A Comprehensive Survey of Research Trends in mmWave Technologies for Medical Applications*", Sensors, 2025.

[**Photoacoustics'25**] [Medical AI] Chuqin Huang, Yanda Cheng, **Xiaoyu Zhang**, Ye Zhan, Wenhan Zheng, Isabel Komornicki, Linda M. Harris, Wenyao Xu, Jun Xia, "*Radiomics-Driven Perfusion Prediction in Clinical Photoacoustic Foot Imaging*", Photoacoustics, 2025.

[**Photoacoustics'25**] [Medical AI] Chuqin Huang, Emily Zheng, Wenhan Zheng, Huijuan Zhang, Yanda Cheng, **Xiaoyu Zhang**, Varun Shijo, Robert W. Bing, Isabel Komornicki, Linda M. Harris, Ermelinda Bonaccio, Kazuaki Takabe, Emma Zhang, Wenyao Xu, Jun Xia, "*Enhanced clinical photoacoustic vascular imaging through a skin localization network and adaptive weighting*", Photoacoustics, 2025.

[**JBHI'24**] [Medical AI] Wei Bo, Suzanne S. Sullivan, **Xiaoyu Zhang**, Mingchen Gao, Wenyao Xu, "*A Telemedicine Analytic Framework for Fully and Semi-automatic Alzheimer's Disease Screening using Clock Drawing Test*", IEEE Journal of Biomedical and Health Informatics, 2024.

[**UIST'23**] [Wireless Sensing] Tiantian Liu, Feng Lin, Chao Wang, Chenhan Xu, **Xiaoyu Zhang**, Zhengxiong Li, Wenyao Xu, Ming-Chun Huang, Kui Ren, "*WavoID: Robust and Secure Multi-modal User Identification via mmWave-voice Mechanism*", ACM Symposium on User Interface Software and Technology, San Francisco, USA, 2023.

[**BSN'19**] [Wireless Sensing] **Xiaoyu Zhang**, Bin Liu, "*A Channel Hopping Strategy Based on the Human Trajectory Similarity for WBANs*", IEEE-EMBS International Conference on Body Sensor Networks, Chicago, USA, 2019.

[**Bodynets'17**] [Wireless Sensing] Chengjie Guan, Bin Liu, Zhiqiang Liu, Y. Zhang, **Xiaoyu Zhang**, "*JMMM: A Mobility Model for WBANs Based on Human Joint Movements*", 19th EAI International Conference on Body Area Networks, Dalian, China, 2017.

Patents

4 patents filed (US). Each corresponds to a first-authored system paper or a deployed prototype.

- **Device and Method for Wound Monitoring and Diagnosis using Radio-Frequency Technologies**. *Corresponds to mmSkin (IoTJ'25).* Under commercialization by a medical-device startup.
- **Autonomous Robotic System for Non-Contact Volumetric Thermal Conductivity Mapping Using Millimeter-Wave Synthetic Aperture Radar**. *Corresponds to IROS'26.*
- **Vision-Based Closed-Loop Control System for Autonomous Biomass Drying with Predictive Crack Prevention**
- **A New mmWave System for Accurate Hand and Finger Tracking**. *Corresponds to mmHand (IoTJ'25).*

Honors and Awards

- **Students and Young Professionals Paper Assistance (SYPA) Award**, IEEE Industrial Electronics Society (IES), for presenting my first-authored paper at IEEE/RSJ IROS 2026
- **Best Paper Candidate Award**, HumanSys'25. The 3rd International Workshop on Human-Centered Sensing, Modeling, and Intelligent Systems (co-located with ACM SenSys), 2025
- **Semifinalist**, Panasci Technology Entrepreneurship Competition (Panasci TEC), University at Buffalo, 2025
- **Graduate Teaching Award**, Department of CSE, University at Buffalo, 2024
- **Chair's Fellowship**, Department of CSE, University at Buffalo, 2021
- Merit scholarships (graduate and undergraduate), USTC and HFUT, 2015–2019

Industry & Clinical Experience

Clinical Research Collaborations *2022 - Present*

Multi-year IRB-approved clinical collaborations at the University at Buffalo and partner sites, spanning device deployment, on-site data collection, and multi-modal cohort curation:

- **Wound care** (UBMD Vascular Surgery). Designed, built and deployed a mmWave clinical wound-imaging station on site; **5 patients** scanned to date under an IRB-approved protocol, with repeat longitudinal scanning up to **3 months** per patient
- **Breast imaging.** Clinical cohort of **284 imaged patients**; curated the **53-patient** tri-modal photoacoustic / ultrasound / mammography subset used for molecular subtyping
- **Wearable ear-EEG. 27 subjects** recorded across four passive auditory tasks (∼1,800 trials), **23** retained after per-channel quality control, paired with standardized clinical score scales
- **Thyroid ultrasound.** Ongoing clinical scanning collaboration for nodule diagnosis and lesion segmentation

- **Photoacoustic Imaging Lab** (Dept. of Biomedical Engineering). Joint imaging-system development and radiomics analysis

China Merchants Bank Software Center, Hangzhou, Zhejiang, China *Jul. 2020 - Sep. 2020*
Designed and implemented a full-stack data-verification system (Bootstrap + Django) with dynamic front-back interaction for multi-source financial data.

Competition

- **Panasci Technology Entrepreneurship Competition (Panasci TEC) 2025** *Mar. 2025*
  Advanced Through-Gauze Wound Assessment Technology.
  Xiaoyu Zhang (technical lead and presenter), Chuhui Liu, Jun Xia, Linda M. Harris, Wenyao Xu.
  Advanced to the **semifinals**; pitched the patented mmWave wound-assessment platform to clinical and industry judges.

Presentations & Demos

- **Conference Talk & Poster**: IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Pittsburgh, USA Sep. 2026
  Autonomous Robotic Inspection of Building Thermal Insulation via Subsurface mmWave Sensing
- **Ph.D. Dissertation Defense**: *Toward High-Resolution and High-Capacity mmWave Sensing Technology for Medical Applications* Jul. 2026
  University at Buffalo, CSE
- **Live Demonstration**: SUNY Research Expo Mar. 2026
  mmWave clinical wound-imaging station, translational outreach
- **Community Engagement Studio**: UB CTSI Feb. 2026
  Wound-measurement device design review with a 12-member community expert panel
- **Invited Talk**: The 3rd International Workshop on Human-Centered Sensing, Modeling, and Intelligent Systems (HumanSys, co-located with ACM SenSys) May 2025
  Through-dressing Wound Monitoring Based on the mmWave Sensor
- **Pitch Talk**: Panasci Technology Entrepreneurship Competition Mar. 2025
  Advanced Through-Gauze Wound Assessment Technology
- **Invited Talk**: IEEE-EMBS International Conference on Body Sensor Networks May 2019
  A Channel Hopping Strategy Based on the Human Trajectory Similarity for WBANs
- **Invited Talk**: 19th EAI International Conference on Body Area Networks Sep. 2017
  JMMM: A Mobility Model for WBANs Based on Human Joint Movements

Teaching Experiences

Teaching Assistant across **6 semesters** at the University at Buffalo, including one invited guest lecture, plus one semester at USTC.

1. Algorithm Analysis and Design, UB *Fall 2021, Spring 2022, Fall 2022, Fall 2023, Fall 2024*
2. Special Topics, UB. **Invited guest lecture**: *Wireless Signal Processing: Making Sense of the Invisible* *Fall 2023*
3. Mathematical Logic and Graph Theory, USTC *Fall 2018*

Mentoring Experiences

I have mentored **10+ students** on wireless sensing, embedded systems, and AI-based medical screening projects. **Four high-school students I mentored are the first and second authors of accepted peer-reviewed papers** (IEEE HealthCom'26 and ACM HumanSys'26).

- Prakriti Setlur, AI-based clinical image screening pipelines
- Kh Shafin Farhan, foundation models for medical imaging
- Abhi Ramtel (CSE@UB), UI design of an over-gauze wound assessment system
- Cole Desimone (AE@UB), 3D model design of the mmWave sensor scanning system

- George Gillman (EE@UB), mmWave technologies for medical applications: a review
- Weida Jiang (CSE@UB), embedded mmWave-based hand detection on Raspberry Pi
- Wenxuan Huang & Yiwen Tan (CSE@UB), tooling for automatic image labeling
- Nicholas Wei & Austin Xu (high school), anatomic-site domain shift in melanoma classification *(first and second authors, IEEE HealthCom'26, accepted)*
- Henry Yang & Emily Liu (high school), real-world study of sport-based support for autism spectrum disorder *(first and second authors, ACM HumanSys'26, accepted)*

Community Services & Outreach Activities

Conference Organization:

- **Organizing Committee Member (Local Arrangements)**, International Conference on Intelligent Biology and Medicine (ICIBM) [2026]

Journal Reviewer:

- IEEE Transactions on Mobile Computing (TMC) [2025, 2026]
- ACM Transactions on Computing for Healthcare (HEALTH) [2025]
- IEEE Open Journal of Engineering in Medicine and Biology (OJEMB) [2025]
- Scientific Reports [2025]
- Smart Health (SH) [2024, 2025]

Conference Reviewer:

- ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE) [2025]
- IEEE International Conference on Robotics and Automation (ICRA) [2027]
- IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI) [2025, 2026]
- Physics Education Research Conference (PERC), American Association of Physics Teachers (AAPT) [2026]
- IEEE-EMBS International Conference on Body Sensor Networks (BSN) [2023, 2024]

Outreach:

- Live technology demonstration, **SUNY Research Expo**. Translational outreach for the mmWave clinical wound-imaging station [Mar. 2026]
- **Community Engagement Studio**, UB Clinical and Translational Science Institute (CTSI). Convened a **12-member community expert panel** (healthcare providers, clinical trainees, community advocates) for a design review of the wound-measurement device [Feb. 2026]